

Exp No: 30 RTOS TASK SCHEDULING

DATE: 25/08/2026

```
kaviya@KV: ~
GNU nano 8.7.1
#include <systemc.h>
SC_MODULE(rtos_scheduler) {
void task1() {
while(true) {
cout << "Task1 Running at "
<< sc_time_stamp() << endl;
wait(1, SC_SEC);
}
}

void task2() {
while(true) {
cout << "Task2 Running at "
<< sc_time_stamp() << endl;
wait(1, SC_SEC);
}
}

SC_CTOR(rtos_scheduler) {
SC_THREAD(task1);
SC_THREAD(task2);
}
};

int sc_main(int argc, char* argv[]) {
rtos_scheduler obj("Scheduler");
sc_start(6, SC_SEC);
return 0;
}
```

```
kaviya@KV:~$ nano exp30.cpp
```

```
kaviya@KV:~$ g++ -o exp30 exp30.cpp -lsystemc
```

```
kaviya@KV:~$ ./exp30
```

```
SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
```

```
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```
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```

```
Task1 Running at 0 s
```

```
Task2 Running at 0 s
```

```
Task1 Running at 1 s
```

```
Task2 Running at 1 s
```

```
Task1 Running at 2 s
```

```
Task2 Running at 2 s
```

```
Task1 Running at 3 s
```

```
Task2 Running at 3 s
```

```
Task1 Running at 4 s
```

```
Task2 Running at 4 s
```

```
Task1 Running at 5 s
```

```
Task2 Running at 5 s
```